



**BIRLA VISHVAKARMA MAHAVIDYALAYA**  
(An Autonomous Institute)  
**INFORMATION TECHNOLOGY DEPARTMENT**  
**AY: 2026-27 Semester: 01**

**PRACTICAL LIST**

**SUBJECT NAME: OBJECT ORIENTED PROGRAMMING WITH C++ (203IT)**

| No. | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Hours |
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| 1.  | <p>A. Write a program to print name and address.</p> <p>B. Write a program to make a Calculator in C++.</p> <p>C. Write a program to convert temp from Fahrenheit to Celsius unit using the equation: <math>C = (F - 32) / 1.8</math>.</p> <p>D. Write a C++ program to find the sum of individual digits of a positive integer.</p> <p>E. Write a program to enter mark of 6 different subjects and find out the total mark. (Using <b>cin</b> and <b>cout</b> statement)</p> <p>F. Write a program to find whether given number is Armstrong or not.</p> <p>G. Write a C++ program to generate all the prime numbers between 1 and n, where n is a value supplied by the user.</p> <p>H. Write a C++ program to sort a list of numbers in ascending order.</p> <p>I. Write a program which uses Concept of Reference Variable.</p> <p>J. Write an inline function to find largest of three numbers.</p> <p>K. Write a program to find simple interest using inline function.</p>                                                                                                     | 2     |
| 2.  | <p>A. Write a program to find area of circle using formula <math>\pi * r * r</math>. Take the value of <b>pi</b> as default argument in function.</p> <p>B. Write a program to swap the value of variables using the concept of call by value and call by reference.</p> <p>C. Write a C++ program to compute area of right angle triangle, equilateral triangle, isosceles triangle using function overloading concept.</p> <p>D. Write a program to overload the max function: (1) Find the max number from three numbers. (2) Find the max string from three strings.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2     |
| 3.  | <p>A. Create a class called 'Student' that has Student_Name, Student_Id, Student_CPI, Student_SPI and Student_Csem as data members. Use appropriate member functions for taking and displaying student information.</p> <p>After creating class below mentioned operations:</p> <ol style="list-style-type: none"><li>Find out details of all those students whose <math>CPI \geq 7.6</math> and <math>\leq 8.9</math></li><li>Display student's details in ascending order of SPI.</li><li>Find out details of user given student ID.</li></ol> <p>B. Define a class to represent a bank account which includes the following members as Data members: <b>a)</b> Name of the depositor <b>b)</b> Account Number <b>c)</b> Withdrawal amount <b>d)</b> Balance amount in the account.</p> <p>Member Functions: <b>a)</b> To assign initial values <b>b)</b> To deposit an amount <b>c)</b> To withdraw an amount after checking the balance <b>d)</b> To display name and balance.</p> <p>Write the above program for handling n number of account holders using array of objects.</p> | 2     |

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| 4. | <p><b>A.</b> Write a program to swap the values of two integer members of different classes using friend function.</p> <p><b>B.</b> Write a program for addition of two complex numbers using friend function (use constructor function to initialize data members of complex class).</p> <p><b>C.</b> Write a program to find the mean value of a given number using friend function.</p> <p><b>D.</b> Write a program to create a class Employee with private data members name and salary to store employee details.</p> <ul style="list-style-type: none"> <li>• Use a suitable function to initialize these data members.</li> <li>• Create a member function display() to show employee details.</li> <li>• Declare a friend function compareSalary() that takes two Employee objects as arguments and compare the salaries of both employees.</li> <li>• Display the name of the employee who has the higher salary, or display a message if both salaries are equal.</li> </ul>                                                                                                                                                                               | 2 |
| 5. | <p><b>A.</b> Write a program to add integer value of two class objects.</p> <p><b>B.</b> Write a program to do addition of two Times objects.</p> <p><b>C.</b> Write a program to design a class having static member function named showCount() which has the property of displaying the number of objects created of the class.</p> <p><b>D.</b> Write a program to check whether given number is prime or not using constructor.</p> <p><b>E.</b> Write a program to calculate factorial of a given number using copy constructor.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2 |
| 6. | <p><b>A.</b> Write a C++ program to demonstrate unary operator overloading. Create a class containing an integer data member and overload the ++ and -- operators. Apply the overloaded operators on an object of the class and display the value before and after increment and decrement operations.</p> <p><b>B.</b> Write a program which uses Concept of Binary Operator Overloading on String. (with and without friend function)</p> <p><b>C.</b> Write a program to create a Matrix and Overload the '+' operator to add two MATRIX objects and '*' object to multiply two matrix.</p> <p><b>D.</b> Construct a class Complex to overload ++ in prefix and postfix version for the complex class. Prefix add to the real part and Postfix add to the imaginary part. For example, if ComplexNo has values 2 + 3i, then ComplexNo++ will make it 2+4i and ++ComplexNo makes 3+3i. Also define appropriate constructor(s) for the same.</p> <p><b>E.</b> Define two classes polar and rectangle to represent points in the polar and rectangle systems. Use conversion routines to convert from one system to the other.</p>                                    | 4 |
| 7. | <p><b>A.</b> Write a program to multiply the positive numbers using single inheritance.</p> <p><b>B.</b> Write a program to multiply of the positive numbers using multiple inheritances.</p> <p><b>C.</b> Consider an example of declaring the examination result. Design three classes: student, exam and result. The student has data members such as rollno, name. Create the class exam by inheriting the student class. The exam class adds data members representing the marks scored in 5 subjects. Derive the result from exam-class and it has own data members like total, average. Write the interactive program into model this relationship.</p> <p><b>D.</b> Write a program in which Vehicle class is defined as base class with price and year of manufacturing. Derive two classes namely Bus and Truck from base class with Bus with seating capacity and price and Truck with loading capacity and price. Develop classes with necessary constructor and member functions calculate to count total price of total seat and calculate total price of item after loading into the truck and display the result use concept of virtual function.</p> | 2 |

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| 8.  | <p><b>A.</b> Create a base class called shape, Use this class to store two double type values that can be used to compute the area of figures. Derive two specific classes called triangle and rectangle from the base shape. Add to the base class, a member function <code>getdata()</code> to initialize base class data members and another member function <code>display_area()</code> to compute and display area of figures. Make <code>display_area()</code> as a virtual function and redefine the function in the derived class to suit their requirements. Using these three classes, design a program that will accept dimensions of a triangle or a rectangle interactively and display area.</p> <p><b>B.</b> Modify the above program to incorporate following changes: a) Make shape class as abstract class with <code>display_area()</code> as pure virtual function b) Use constructor function to initialize the data members of base class not through the <code>getdata()</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                        | 2 |
| 9.  | <p><b>A.</b> Write a program to read a list containing item name, item code and cost interactively and display the data in a tabular format.</p> <p><b>B.</b> Write a program to design your own manipulator to provide the output specification for printing money value.</p> <p><b>C.</b> Write a program to swap the numbers using the concept of function template.</p> <p><b>D.</b> Write a program to create class template to find the greater of the given two numbers.</p> <p><b>E.</b> Declare a template class called <code>MyArray</code> having an array of generic type as a data member, named <code>elements[10]</code>.<br/>Define following generic (template) member functions:<br/>- sort to arrange elements in ascending order<br/>- <code>find_max</code> to find and return maximum from the array<br/>Define main to illustrate usage of these functions to process two different types of data.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2 |
| 10. | <p><b>A.</b> Write a program to copy content of one file to another file.</p> <p><b>B.</b> Write a program which write and read object of Student class from binary file using <code>write()</code> and <code>read()</code> functions. Student class has <code>Student_Name</code>, <code>Student_id</code>, and <code>Student_CPI</code> as data members. The <code>Student_Name</code> and <code>Student_id</code> should be left justified and the <code>Student_CPI</code> should be right justified with two precision value.</p> <p><b>C.</b> Write an interactive program to compute square root of a number. The input value must be tested for validity. If it is negative, the user defined function <code>my_sqrt()</code> should raise an exception.</p> <p><b>D.</b> Write a C++ program to create a class Student which have Student Name, Student Department, CPI and SPI as data members. Take N Number of Student Information and store it into "StudentDB.txt" file. Also read the same file and display the information in format mentioned herewith. Student name left justified with width of 30. For CPI and SPI make user defined manipulator which contains width of 6, precision of 2 and show trailing zero. Student Department must be right justified with width of 25. During reading of the file, if CPI and SPI of any student found same then generate user defined exception.</p> | 2 |

**Subject Coordinator:** Dr. N. B. Prajapati